

PAYFINIA • MONEY MOVEMENT ANALYTICS

# The ROI of Instant Payments

What a community financial institution gains by migrating from checks and wires to FedNow & RTP

**Deliverables 1 & 2 • Public Data Synthesis + Migration ROI Model**

USF FinTech Graduate Project Team • Prepared for Payfinia

# One question every CFI asks about instant payments

*“If I move transaction volume from legacy rails — checks, wires, Same-Day ACH — onto instant rails (FedNow, RTP), what is the financial return?”*

Today that question is answered anecdotally. Our job: replace the guess with a transparent, defensible, data-driven model.

## Legacy rails (today)

Checks, wires, Same-Day ACH — high cost, slow, fraud-prone

## Instant rails (the shift)

FedNow & RTP — near-instant, low fee, 24/7

## The gap = the opportunity

Every dollar of legacy cost displaced is margin for the CFI

# What we built — four deliverables

1

## Public Data Synthesis

A sourced reference base of rail volumes, costs, fraud & global adoption

DONE

2

## Migration ROI Model

A full-cost model (Python + Excel) quantifying migration savings

DONE

3

## CFI Archetype Analysis

Clustering banks into types mapped to ROI ranges

NEXT

4

## Interactive Calculator

A tool the go-to-market team uses with prospects

PLANNED



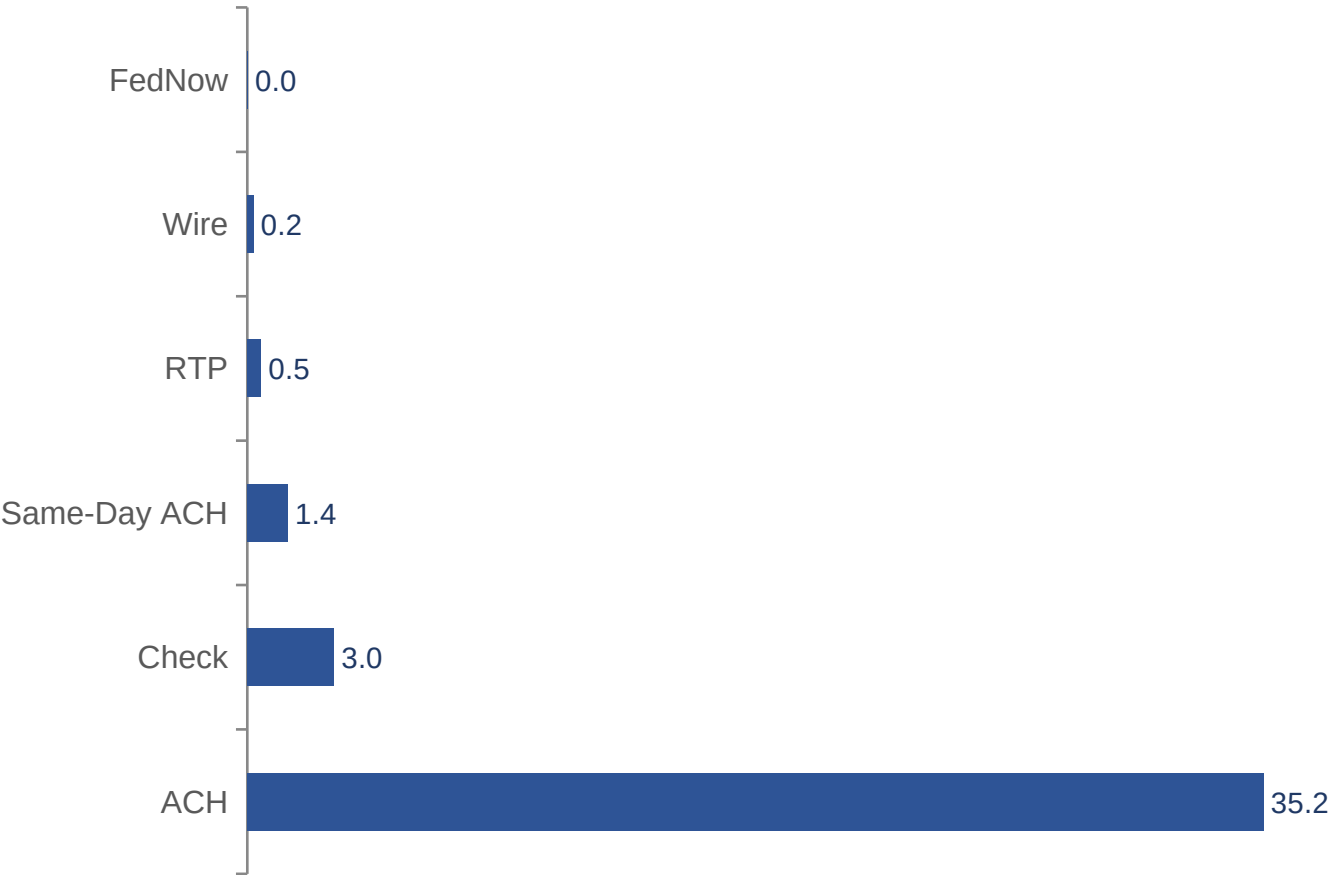
# 1

PART 1

## The Data

What the public record says about payment rails — every figure sourced

# Legacy rails still carry almost all the volume

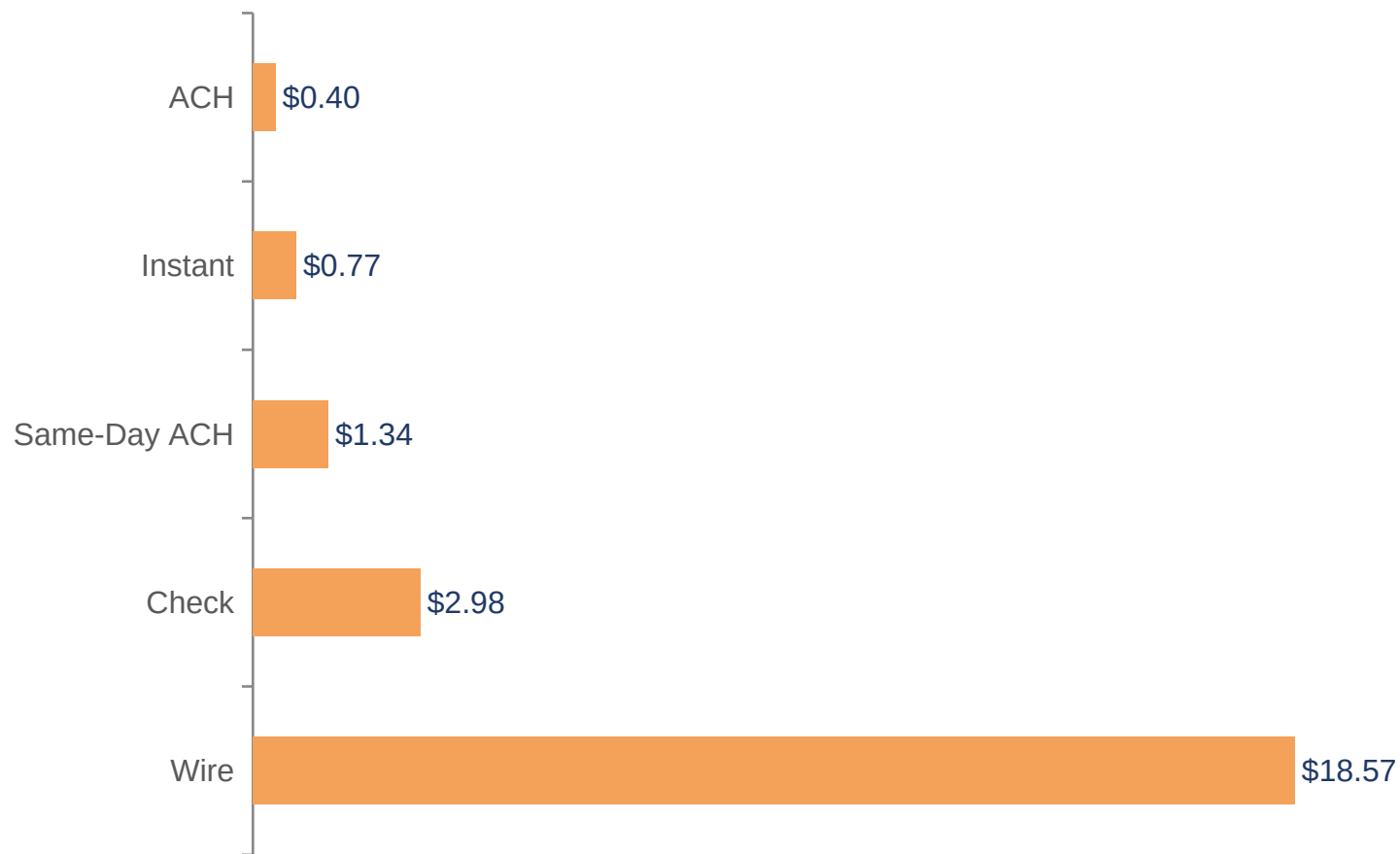


## Read this

**ACH alone is 35 billion payments a year.**  
Instant rails (FedNow, RTP) are tiny today.

That gap is the opportunity — enormous legacy volume is still available to migrate.

# Why migration pays — cost to move one transaction



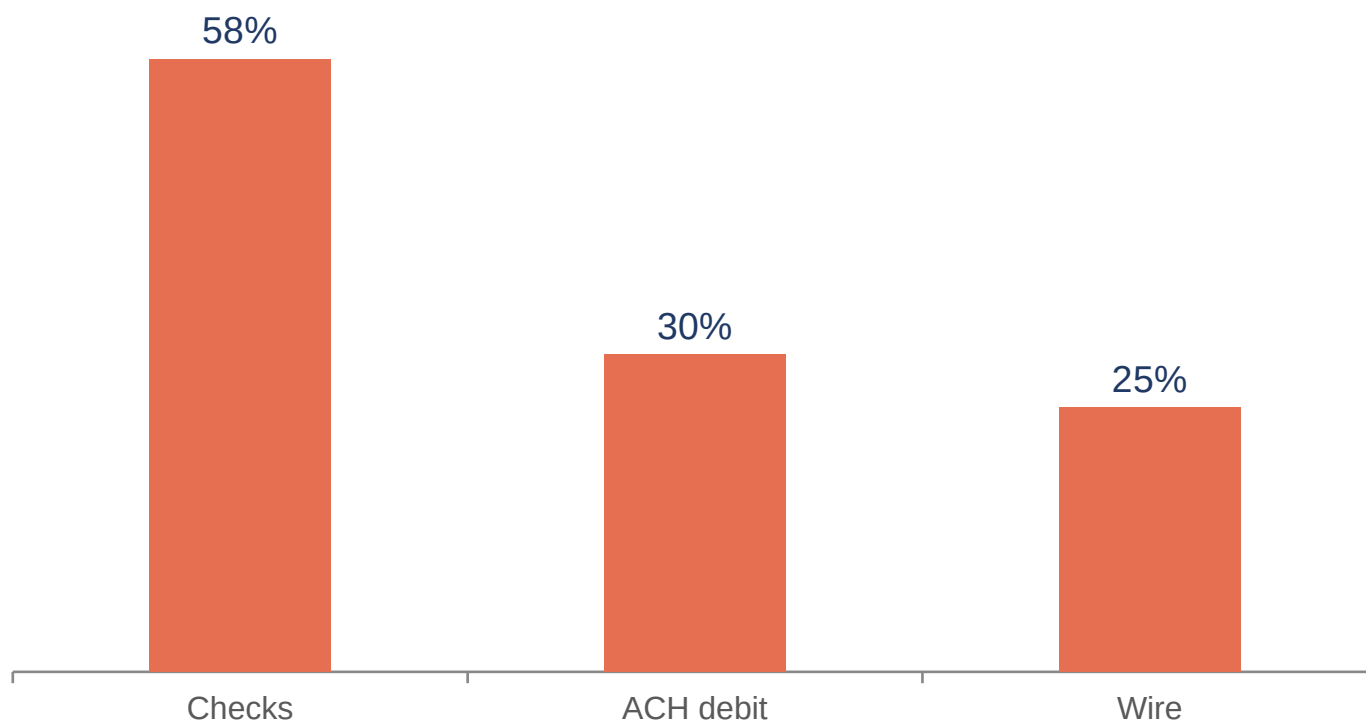
## The whole case

A wire costs \$18.57 and a check \$2.98 to process — an instant payment ~\$0.77.

**Standard ACH is already cheap, so moving it saves nothing.**

**The savings come from displacing checks and wires.**

# Legacy rails carry the most fraud



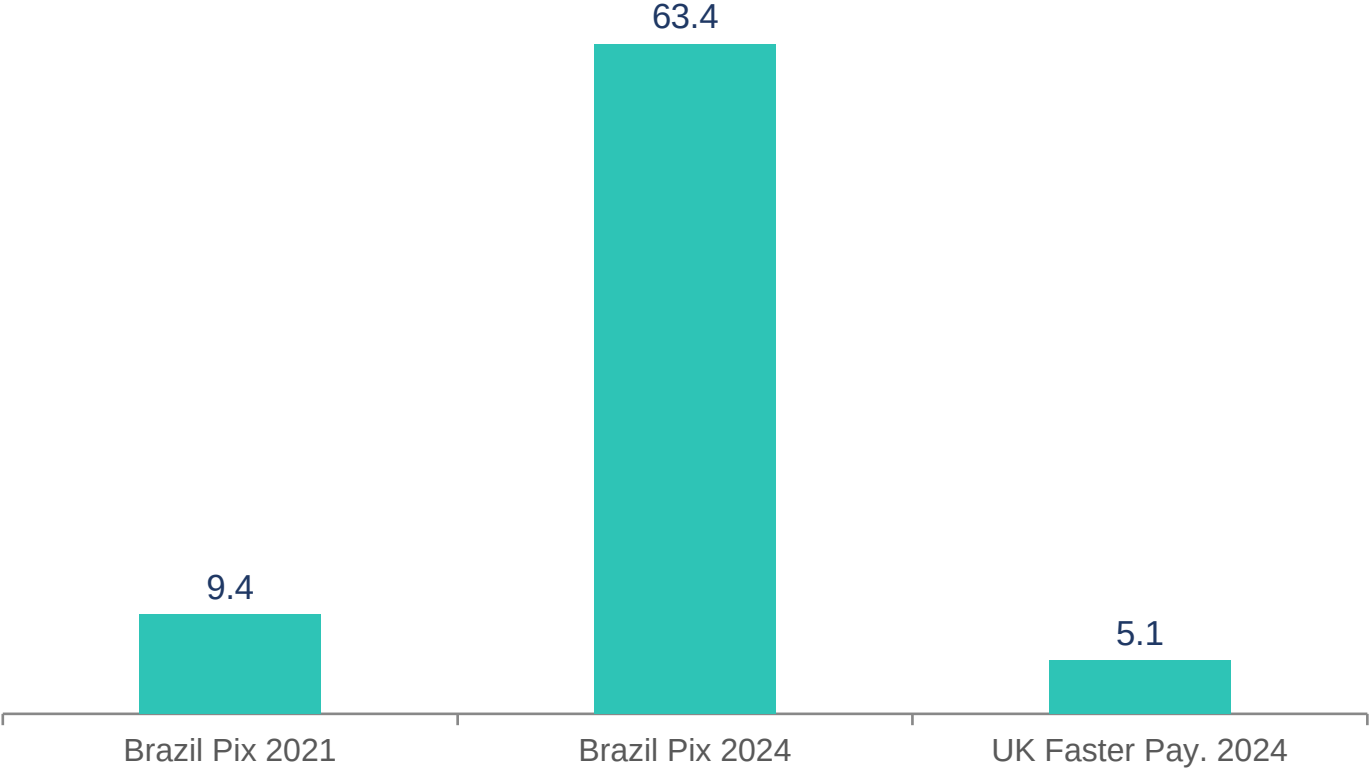
## Read this

Checks are the most-targeted instrument — 58% of firms hit in 2025.

**Displacing checks cuts cost AND exposure.**

*Caveat: instant payments are irrevocable — they need strong pre-send verification.*

# It already happened abroad



## The analog

Brazil's Pix went from 9.4B to 63.4B transactions in three years and pushed checks to near zero.

The UK shows the same shift at a steadier pace.

**These bracket how fast U.S. migration can go.**



# 2

PART 2

## Unit Economics of Each Rail

We rebuilt each rail's cost from 8 components — no lumped numbers

# Every rail's cost is built from 8 components

1

## Network fee

Price paid to the rail operator

2

## Processing / operations

Internal labor & systems to move the item

3

## Failure & exception

Returns, rejects, repairs (rate × cost)

4

## Fraud loss

Expected fraud write-off per item

5

## Fraud prevention

Screening & monitoring tooling

6

## Compliance

AML / sanctions screening per item

7

## Reconciliation

Matching & breaks handling

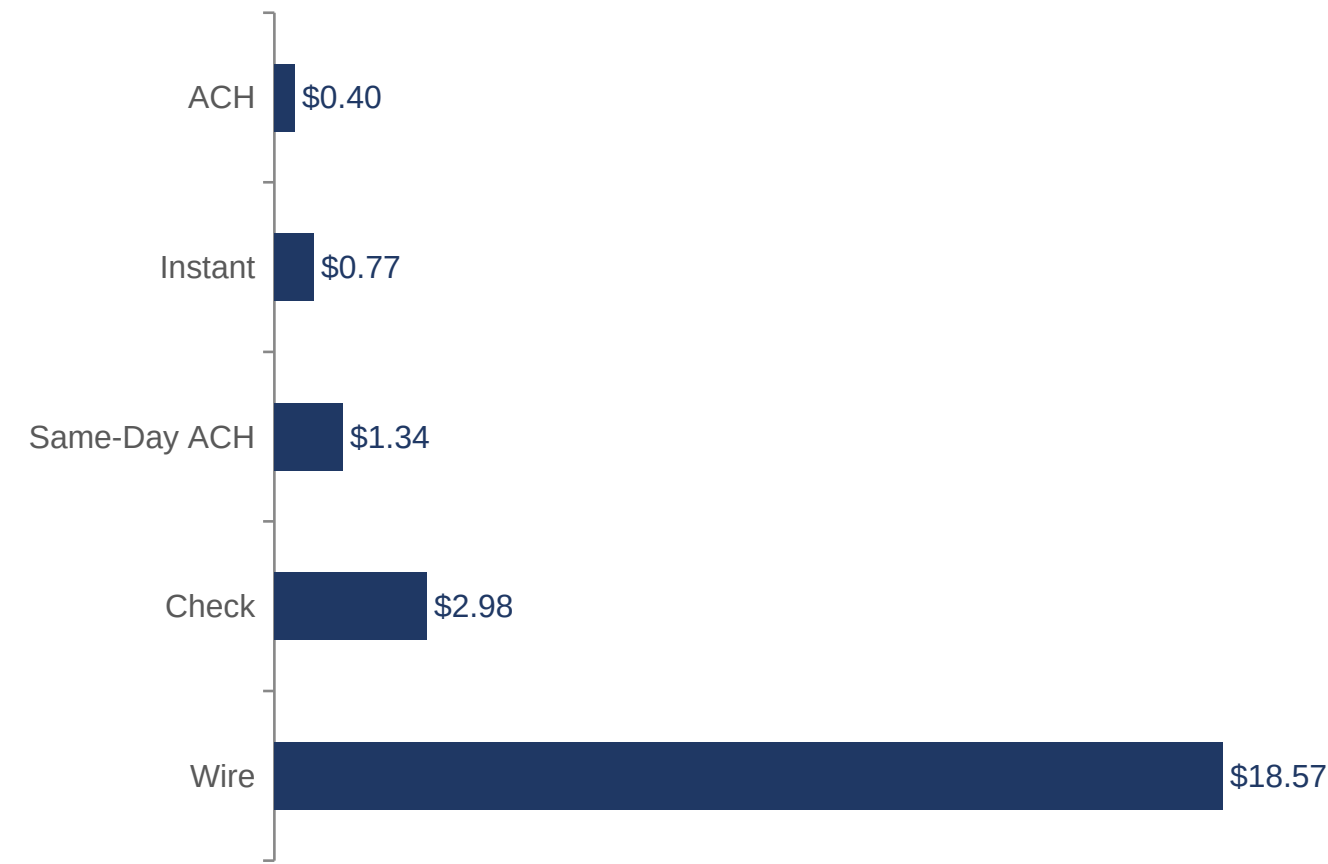
8

## Liquidity / prefunding

Funded-position carrying cost (instant)

*These eight sum to the fully-loaded cost per transaction. Every value is sourced and adjustable.*

# The five rails, side by side

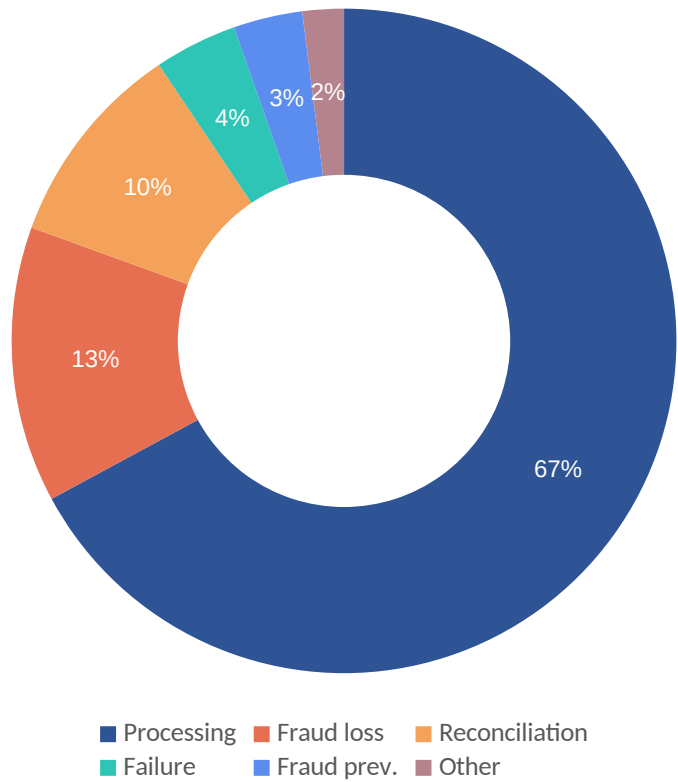


| Rail         | Cost/txn | Save vs instant |
|--------------|----------|-----------------|
| Check        | \$2.98   | +\$2.21         |
| Wire         | \$18.57  | +\$17.80        |
| Same-Day ACH | \$1.34   | +\$0.57         |
| ACH          | \$0.40   | -\$0.38         |
| Instant      | \$0.77   | baseline        |

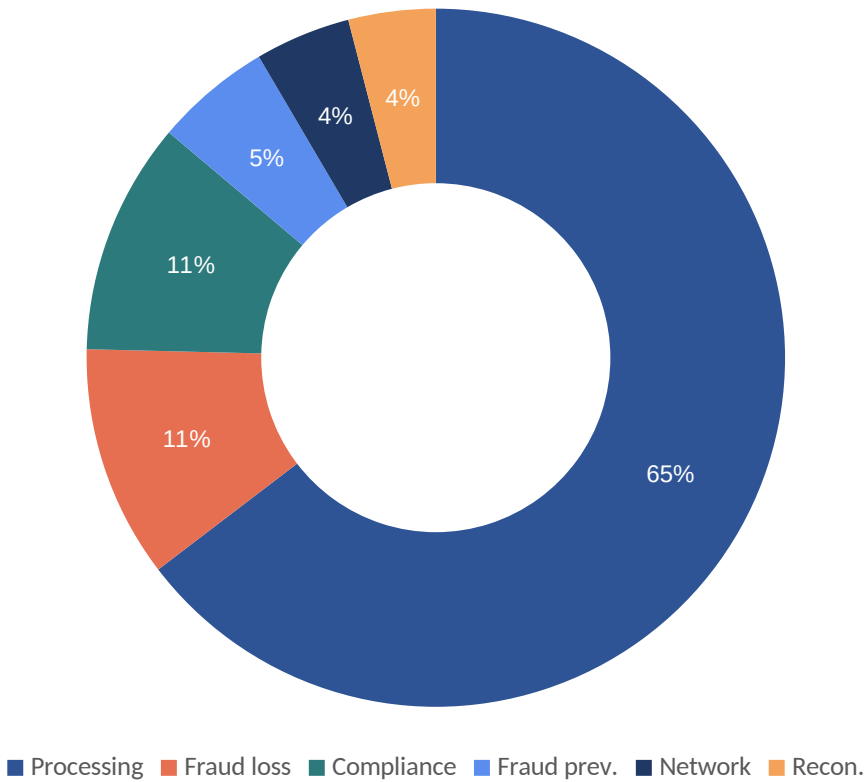
ACH shows a negative saving — migrating it costs more. That's an honest result.

# Checks & wires — where the cost actually sits

CHECK · \$2.98 / txn



WIRE · \$18.57 / txn



## What the detail exposed

### Instant is NOT free

**\$0.77**

fully-loaded cost per instant txn

Real-time fraud screening, irrevocable-payment (APP scam) loss risk, and 24/7 prefunding add real cost — but still a fraction of checks & wires.

### Migrating ACH loses money

**-\$0.38**

cost per ACH txn if migrated

Standard ACH already costs ~\$0.40 — less than instant's full \$0.77. So we do NOT count ACH migration as a saving. The ROI is a check-and-wire story.



# 3

PART 3

## The ROI Model

From cost data to a dollar answer — with sensitivity and honest caveats

# Four transparent steps

## 1 Cost structure

8-part fully-loaded cost per rail (network → liquidity)

## 2 Substitution

Only a share of each legacy rail's volume migrates — never 100%

## 3 Savings

Legacy cost minus instant cost, times migrated volume, per rail

## 4 Returns

Net benefit → ROI, payback period, and 5-year NPV

RESULTS

# Estimated annual impact across three CFIs

\$43K

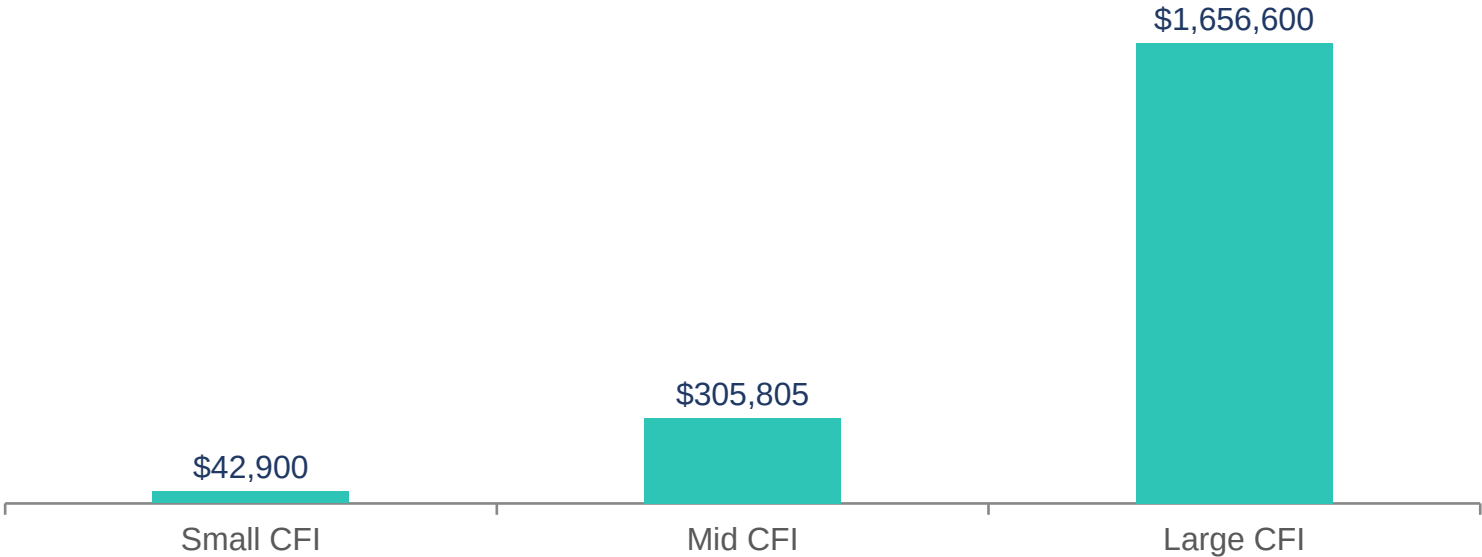
Small CFI • net / yr • 71% ROI

\$306K

Mid CFI • net / yr • 204% ROI

\$1.66M

Large CFI • net / yr • 473% ROI



**Payback**

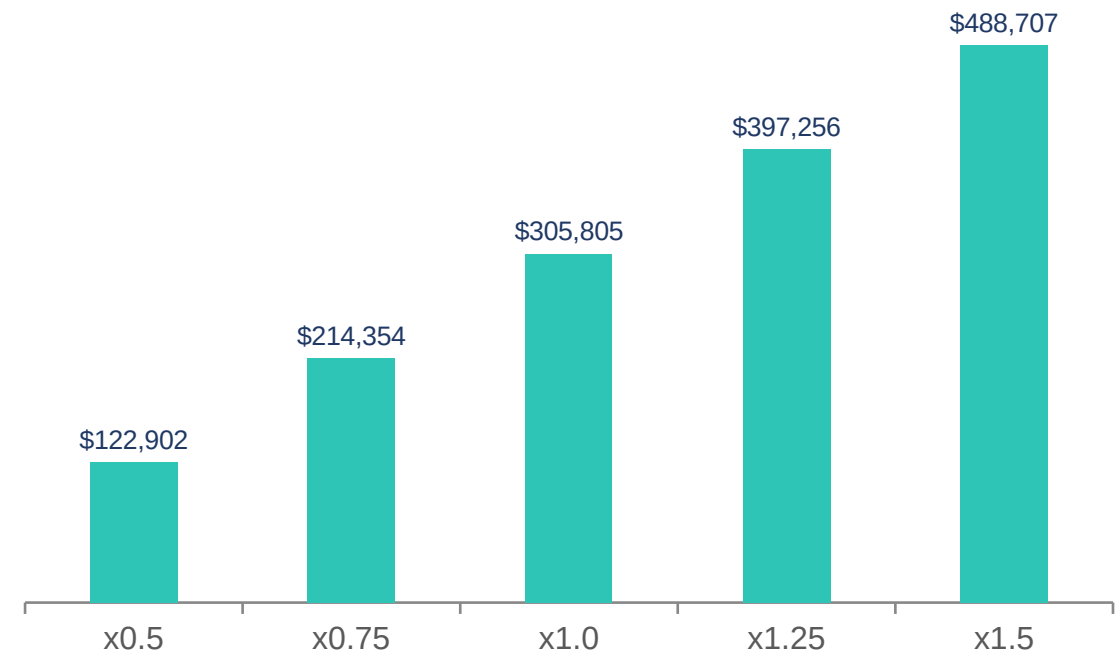
Small 16.8 mo • Mid 5.9 mo • Large 2.5 mo

**5-year NPV (@10%)**

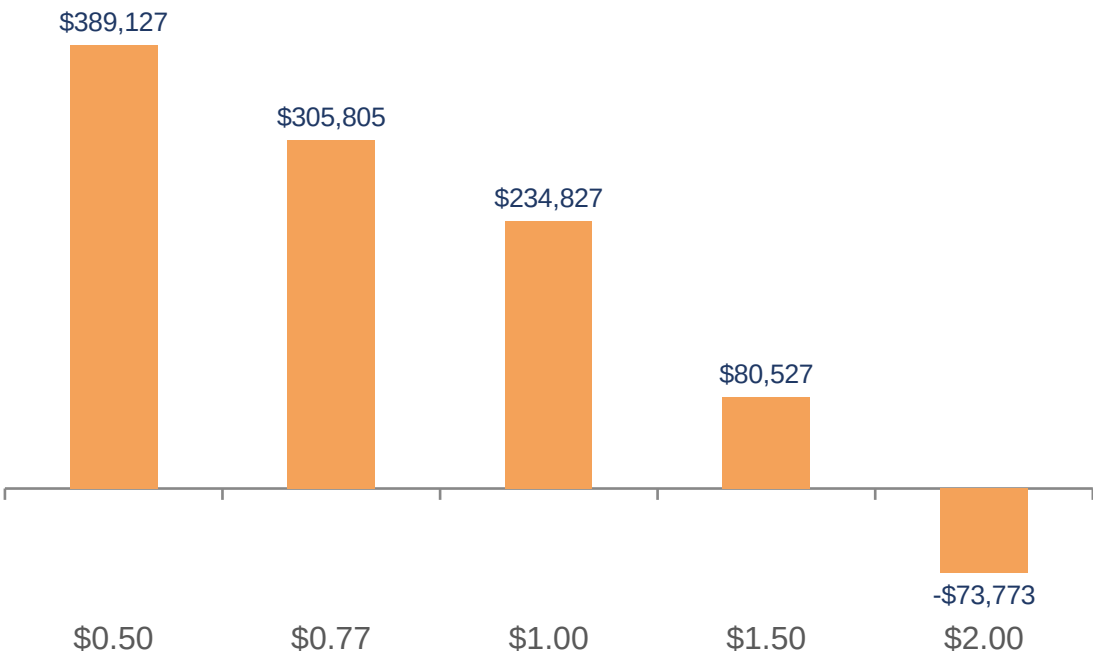
\$0.1M • \$1.0M • \$5.9M

# What actually drives the ROI (Mid CFI)

If check & wire migration is HALF our estimate



If instant's true cost is higher than \$0.77



ROI is most sensitive to how much check/wire volume migrates and to instant's true cost. At ~\$2/txn, migration stops paying — the stress test a sharp prospect will run.

# Illustrative until calibrated with Payfinia data

## **Fraud loss & prevention per item**

Public data gives 'share of firms hit', not \$ per transaction

## **True processing labor per item**

Varies by core processor and institution size

## **Failure / return rates**

Nacha thresholds bound them; actuals differ by CFI

## **Real substitution rates**

International analogs bound the range; production data confirms it

*The structure is correct; these inputs get sharpened with real data. The model is built so Payfinia can plug them in without re-engineering.*

## RECOMMENDATION

# So — what should a CFI do?

### MOVE TO INSTANT

**Checks & wires → FedNow / RTP**

Biggest cost savings and fraud reduction.  
This is where the ROI lives.

### KEEP FOR NOW

**Standard ACH stays**

Already cheap and efficient. Migrating it  
costs more, not less.

### PRIORITIZE BY MIX

**Target high check/wire  
volume**

A CFI's ROI depends on its rail mix. More  
checks & wires = bigger payoff.

WHAT'S NEXT

## Phase 3 — CFI Archetypes

Cluster real banks (public FDIC / NCUA data) into archetypes and map each to an expected ROI range — so Payfinia can tier its prospect pipeline by likely return.

**Takeaway:** the migration case is real, defensible, and driven by checks and wires — calibrate with production data and it's ready for prospect conversations.